

Peer Review File

An explainable longitudinal multi-modal fusion model for predicting neoadjuvant therapy response in women with breast cancer



Open Access This file is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to

the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made. In the cases where the authors are anonymous, such as is the case for the reports of anonymous peer reviewers, author attribution should be to 'Anonymous Referee' followed by a clear attribution to the source work. The images or other third party material in this file are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this license, visit <http://creativecommons.org/licenses/by/4.0/>.

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author): expertise in multi-modal model development

Gao et al present a multimodal framework for predicting response to neoadjuvant chemotherapy. The model is unique in that there are 2 modules, consisting of a model that analyzes pre-NAT mammograms and another that uses longitudinal MRI which can include scans at pre/mid/post-NAT. Both models leverage radiological, histopathological and clinical data. The modular design of the framework is novel, as well as the method of training via cross-modal knowledge mining. The ablation analysis is also appropriate, evaluating the modules in-silo and without the cross-modal knowledge predictor.

However, improvements could be made in presentation to make the manuscript more concise and better communicate the results.

1. There are too many display items (8 figures, 5 tables) which is over the maximum 10. I would suggest moving Figure 3, 4 and 6 to supplement. Figure 7 and 8 could be combined and also moved to supplement. Tables 1-4 could be combined into a single characteristics table with a condensed number of rows. Table 5 could be placed in supplement. Additionally, Figure 2 could benefit from removing b and c and instead using the space to provide more detail in describing the model. Please note that these are all merely suggestions and the authors should condense their display items in the way they feel best communicate their results.
2. There should not be a "Conclusion" section.
3. Can the uni-modal rhpc model be explicitly defined?
4. L122-143 does not need AUROC comparisons between MRP and each reader individually. A summary is sufficient.
5. The description of the model could benefit from additional explanations. There should be clarification on how the rhpc variables are used as model targets but also included in the model itself via the MLP. It is also unclear how the temporal information is encoded. How does the linear layer + ELU construction allow for temporal representations?
6. Figure 5 indicates that much of the power may come from rhpc variables, is there a comparison to an rhpc-only model?
7. This work has focused on deriving data-driven features from radiological imaging and fusing these features with other characteristics (rhpc). The discussion should reference progress in using data-driven features from histological data for prediction of NAT which in the future could be fused with imaging features which could further improve performance.

Reviewer #2 (Remarks to the Author): clinical expertise in breast cancer therapy response

In this research, Gao and colleagues embarked on a pioneering project to develop a Multi-modal Response Prediction (MRP) for early-stage breast cancer, leveraging clinical-pathological and radiological imaging data. The endeavor to identify predictors of pathological complete response (pCR) is a pivotal area of interest in breast cancer research. To date, numerous biomarkers have been scrutinized for their predictive value using pre-treatment baseline characteristics across various breast cancer subtypes and therapeutic approaches. The

significance of pCR status is well-established in HER2-positive and triple-negative breast cancer due to its profound correlation with enhanced long-term survival. However, its relevance in HR+/HER2- breast cancer remains ambiguous. Additionally, the specific therapeutic agents achieving pCR are critical, as they bear implications for long-term outcomes.

While the authors have undoubtedly invested a considerable effort in this study, meriting recognition, I harbor significant reservations regarding the originality of their findings and their impact on the breast cancer domain. My concerns are as follows:

The bulk of the predictive outcomes and their comparisons with interpretations by radiologists, as well as with other clinical-pathological indicators, are derived from a substantial training dataset.

The performance metrics within the genuinely independent dataset (i.e., the Duke cohort) leave much to be desired. Furthermore, the sample size of this independent dataset is notably small, particularly for certain subsets such as HER2+ or those receiving endocrine therapy, complicating the reliability of any derived conclusions.

The models used prior to neoadjuvant therapy (pre-NAT) appear to yield results comparable to those of other models that consider imaging during and after treatment. This raises the question of whether these pre-NAT models surpass existing predictive biomarkers like OncotypeDX, HER2DX, or the model recently introduced by Carlos Caldas's team in *Nature Cancer* (<https://www.nature.com/articles/s41586-021-04278-5>).

The use of on-treatment MRIs is not uniformly accepted worldwide nor endorsed by clinical guidelines.

The authors' conclusion that the MRP could "reduce unnecessary treatment toxicity in non-pCR cases by 35.8% without halting effective treatment in the Pre-NAT scenario, and potentially eliminate surgery for pCR cases by 16.7% without foregoing treatment opportunities in the Post-NAT scenario" is, in my view, flawed. This is because there is no established efficacy for altering therapy based on an anticipated lack of pCR. Moreover, forgoing surgery is not a current standard of care, and the necessity for prospective trials (some of which are currently underway) renders this conclusion somewhat misleading.

Given these points, while the study represents a significant undertaking, its contributions to the field of breast cancer, particularly in terms of novelty and clinical applicability, require a more critical examination.

Reviewer #3 (Remarks to the Author): expert in breast cancer radiology

NCOMMS-24-02259

Summary

In this retrospective study, the authors develop and validate an explainable longitudinal multi-modal fusion model, the Multi-modal Response Prediction (MRP) system, to predict NAT response in women with breast cancer. The model was developed and tested using data from 3,352 patients with breast cancer who were treated with NAT in the Netherlands and externally

tested on a dataset from Duke University. The MRP is composed of two independently trained models: one that includes pre-NAT mammographic imaging and radiological, histopathological, personal, and clinical variables and one that includes longitudinal MR imaging and the aforementioned variables. The MRP system achieved an AUROC of 0.882 for pCR prediction prior to the initiation of NAT and an AUROC of 0.889 for having achieved pCR during and after NAT. The MRP system also outperformed radiologist readers in pCR prediction prior to the initiation of NAT.

Strengths

- A. This topic is of high clinical relevance and importance – if NAT response can be predicted with accuracy, then NAT regimens could be modified early on based on patient response and potential toxicities.
- B. The MRP system integrates multi-modal data from a large cohort of breast cancer patients who underwent NAT (n=3,352).
- C. The authors also conducted a reader study and an error analysis to better understand discrepancies between MRP and radiologist predictions.
- D. The tables and figures are detailed and comprehensive.

Weaknesses

- 1. The MRP system was validated on an external dataset from Duke University; however, that external dataset included only pre-NAT MRI examinations (no other MRI examinations or pre-NAT mammographic images), and thus the MRP system was not comprehensively tested.
- 2. Clarification about the ground truth reference standard should be provided – in particular, the “unified pCR standard” should be defined and whether the presence of DCIS (but not invasive disease) is considered pCR.
- 3. The authors should clarify how the 120 patients for test cohort A were selected. Since all had mammogram and MRI examinations available for analysis, their selection does not seem random. Furthermore, the authors should verify that all reported AUROC values refer to test set performance.
- 4. Ultrasound and MRI are most commonly used to monitor NAT response. The authors should address in the Discussion section why mammography rather than ultrasound was used in the MRP system.
- 5. More detail and discussion about the modality contribution analysis would be helpful (specific questions below). Figure 5 suggests that radiological features are less important than histopathological and other features.
- 6. In the Methods section, the authors should clarify what criteria are used at their institution to determine whether or not a patient undergoing NAT is monitored with MRI.

Specific Comments

- 7. Title: Please consider “...for predicting neoadjuvant therapy response in women with breast cancer.”
- 8. Abstract: Please consider changing “clinical reality” to “the real-world clinical setting.”
- 9. Abstract: Please consider changing “real clinical pathway” to “pathways used in real-world clinical settings.”
- 10. Abstract: Please spell out pCR upon first use.
- 11. Introduction, line 27: Please clarify what is meant by “radical” surgery.
- 12. Introduction, line 46: Please correct “patient prognosis care.”
- 13. Results, Patient Data: How were the 120 patients for test cohort A selected? Since all had

mammogram and MRI examinations available for analysis, their selection does not seem random.

14. Results, Primary Results, line 100: Please clarify what is meant by “During the Pre-NAT period” – does this refer to pre-NAT mammographic and MR imaging?

15. Results, Primary Results, line 103: Please clarify what is meant by the uni-modal rhpc model.

16. Results, Primary Results, lines 105-106: Given that the improvement is not statistically significant, please consider tempering the statement that “the MRP’s performance surpasses that of the Pre-NAT period.”

17. Results, Primary Results, line 122: The paragraph beginning with “MRP outperforms six international breast radiologists...” is difficult to follow.

18. Results, Modality Contribution Analysis, lines 149-150: What is meant by in situ status (invasive + in situ disease)? What is meant by density in this context (mammographic breast density)?

19. Results, Modality Contribution Analysis: With regard to the top five significant features, what is the direction of the association with pCR?

20. Results, Personalized Management: Please correct subtitle “bilateral caring.”

21. Results, Personalized Management, line 194: Please clarify what is meant by “although radiologists offered a better choice than the standard clinical approach...”

22. Discussion: Ultrasound and MRI are most commonly used to monitor NAT response. Please address why mammography rather than ultrasound was used in the MRP system.

23. Discussion: In my opinion, the primary benefit of the MRP system is its potential use to modify and personalize NAT regimens. The other proposed use – to avoid definitive surgery – could result in adverse patient outcomes if complete pCR is incorrectly predicted by the system.

24. Methods: At the authors' institution, what criteria are used to determine whether or not a patient undergoing NAT is monitored with MRI? Or, is MRI routinely obtained in all patients undergoing NAT?

25. Methods, Inclusion and Ethics Statement, line 394: Relatively few patients (n=120) had both pre-NAT MG and MRI.

26. Methods, Ground Truth, line 413: Please clarify what is meant by a “unified pCR standard” or provide a reference.

27. Figure 1: Please define “TN.”

28. Figure 5: These graphs suggest that radiological features are less important than histopathological and other features. Please comment on this in the Discussion section.

29. Figures: Please consider adding a patient flowchart that details the exclusion criteria.

30. Table 1: To what does location refer (C50.0, etc)? Please clarify.

Reviewer #3 (Remarks on code availability):

I was able to access the algorithm code at the following link: <https://github.com/yawwG/MRP/>.

RESPONSE TO REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author): expertise in multi-modal model development

Gao et al present a multimodal framework for predicting response to neoadjuvant chemotherapy. The model is unique in that there are 2 modules, consisting of a model that analyzes pre-NAT mammograms and another that uses longitudinal MRI which can include scans at pre/mid/post-NAT. Both models leverage radiological, histopathological and clinical data. The modular design of the framework is novel, as well as the method of training via cross-modal knowledge mining. The ablation analysis is also appropriate, evaluating the modules in-silo and without the cross-modal knowledge predictor.

We sincerely thank the referee for recognizing and highlighting the novelty and contributions of our work, along with positive and encouraging comments.

However, improvements could be made in presentation to make the manuscript more concise and better communicate the results.

1. There are too many display items (8 figures, 5 tables) which is over the maximum 10. I would suggest moving Figure 3, 4 and 6 to supplement. Figure 7 and 8 could be combined and also moved to supplement. Tables 1-4 could be combined into a single characteristics table with a condensed number of rows. Table 5 could be placed in supplement. Additionally, Figure 2 could benefit from removing b and c and instead using the space to provide more detail in describing the model. Please note that these are all merely suggestions and the authors should condense their display items in the way they feel best communicate their results.

We agree that the tables and figures in the main section could be more concise and better communicate the results. We have made the following changes:

Figures 3, 4, 6:

- We have moved the original Fig. 4 to supplementary materials (revised Supplementary Fig. 4).
- For Fig. 3, and Fig. 6 (which is now shown in Fig. 5), we placed them in the main section as we feel that it would be more accessible to the primary study results. Following your suggestions, detailed results are now presented in supplementary materials (Supplementary Figs. 2, 3, 19), which we feel is more predominantly be of interest to a specialist readership.

Figures 7, 8:

- We have combined the original Figs. 7 and 8 and moved to supplementary materials (revised Supplementary Fig. 1a and 1b).

Tables 1-4:

- We have integrated original Tabs. 1-4 into Tab. 1.

Table 5:

- We have altered Tab. 5, which is now shown in revised Tab.2 (models' performance) and Supplementary Tab.1 (detailed readers' performance).

Figure 2:

- We have extended Fig. 2a to enhance clarity on the model's structure, detailing the pathways for multi-modal imaging and rhpc-based inputs in representation embedding, cross-model learning, and their integration within each model for neoadjuvant therapy response prediction.

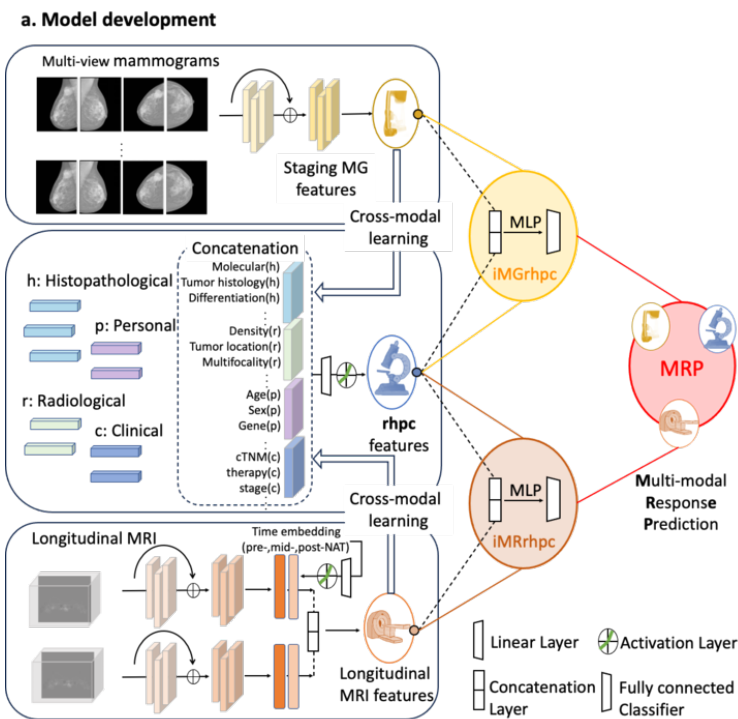


Figure 2a: Amended to address Reviewer 1's comment 1

We have revised the figure legend accordingly:

"Both iMGrhpc and iMRrhpc consist of two modules: one module is for cross-modal knowledge learning that predicts rhpc information using only imaging features, and another module is for response prediction using integrated features of rhpc-based and imaging features."

- We have updated Fig. 2b and 2c, which are now shown in revised Fig. 2c, to directly present the model's performance in a way that is more accessible to a broader audience.

2. There should not be a “Conclusion” section.

We thank the reviewer's remarks. We have removed the “Conclusion” section and added relevant contents to the last paragraph of Discussion section on page 13, lines 402-412.

3. Can the uni-modal rhpc model be explicitly defined?

We agree that the uni-modal rhpc model should be explicitly defined. We have expanded the uni-modal rhpc model definition on page 16, lines 503-508, as follows.

“We first trained the uni-modal rhpc-based model as the baseline model, which extracts features from rhpc characteristics (r: radiological, h: histopathological, p: personal, c: clinical) as presented in Tab.1. Age of patients was modeled as a continuous variable, while all other characteristics were encoded as one-hot categorical variables. These variables were then embedded through a linear layer followed by an Exponential Linear Unit (ELU) activation function¹. Subsequently, the features from all variables were concatenated and fed into a two-layer multi-layer perceptron (MLP) with output dimension of 256. Finally, the MLP output is passed through a linear classifier to estimate the treatment response of breast cancer patients.”

4. L122-143 does not need AUROC comparisons between MRP and each reader individually. A summary is sufficient.

We have shortened the AUROC descriptions with a concise summary, please see page 5, lines 152-157.

“In a comparison between the MRP and six international radiologists' AUROC, significant differences were found in the pre-NAT phase where the MRP outperformed all six readers (Δ AUROC ranging from 0.159 to 0.373, DeLong's test: p -values < 0.001). In the Mid-NAT and Post-NAT phases, the MRP continued to significantly outperform four readers (Mid-NAT: Δ AUROC ranging from 0.106 to 0.166, with p -values between $1.798e-02$ and $3.0823e-03$; Post-NAT: Δ AUROC ranging from 0.093 to 0.140, with p -values between $2.437e-02$ and $4.175e-03$). Further detailed results for each reader's evaluation can be found in Extended Data Figs. 3 and Extended Data Tab.1.”

5. The description of the model could benefit from additional explanations. There should be clarification on how the rhpc variables are used as model targets but also included in the model itself via the MLP. It

is also unclear how the temporal information is encoded. How does the linear layer + ELU construction allow for temporal representations?

We agree with the reviewer that more detail around the proposed model is appropriate.

rhpc variables: When utilizing rhpc variables as model targets, they constituted a distinct branch (i.e., cross-model learning) based only on extracted imaging features. From the multi-task learning point of view², incorporating multiple relevant learning targets can enhance the feature extraction capabilities of the imaging encoder (see Supplementary Figs. 10 and 12-14). When utilizing rhpc variables as inputs for treatment response prediction, we integrated all available information, including imaging and rhpc features. We have revised Fig. 2a accordingly and provided detailed rhpc variables deployment (see comment 1 for details on the revised Fig. 2a).

Temporal information encoding: In this study, temporal information refers to three time points: pre-NAT, mid-NAT, and post-NAT, which are encoded as one-hot categorical variables. The one-hot vectors are then constructed using a linear layer followed by the Exponential Linear Unit (ELU) activation function, a method widely used in previous studies^{3,4}. This approach enables these variables to be embedded as representations, enhancing the capture of interactions between features and optimizing the use of numerical data.

Action taken: We have provided the temporal information encoding details on page 17, lines 521-524.

“Subsequently, the extracted image features are embedded with stage-specific temporal information. This includes three temporal signals for Pre-, Mid-, and Post-NAT, which we converted into one-hot categorical variables. The one-hot vectors are then constructed using a linear layer followed by the Exponential Linear Unit (ELU) activation function, a method widely used in previous studies^{3,4} allowing for the integration of temporal representations.”

6. Figure 5 indicates that much of the power may come from rhpc variables, is there a comparison to an rhpc-only model?

Yes, the comparison results to rhpc-only model were shown in Fig.3. As the reviewer rightfully notes, the inclusion of rhpc features is important to the overall training model. We have conducted a feature importance analysis of the rhpc-only model. The top significant features predicting pCR are aligned with established findings in our study (lines 166-174), including histopathological factors (such as molecular subtype, in situ component status, and tumor histology) and radiological factors (such as tumor multifocality).

Action taken: We have incorporated the feature contribution results of the rhpc model into the original Fig. 5 (now referred to as Fig. 4).

7. This work has focused on deriving data-driven features from radiological imaging and fusing these features with other characteristics (rhpc). The discussion should reference progress in using data-driven features from histological data for prediction of NAT which in the future could be fused with imaging features which could further improve performance.

We appreciate the reviewer's suggestion. In the revised manuscript on page 13, lines 396-401, we have included this point in the discussion section and incorporated references ^{5,6}, discussing the advancements in utilizing data-driven features from histological data for predicting NAT outcomes.

“Fifthly, this study demonstrates the designed MRP system’s efficiency in integrating rhpc data with breast radiological imaging modalities for NAT response prediction. However, our current models were developed solely using two radiological modalities (mammography and MRI). Recent studies ^{5,6}, known to utilize digital pathology data with whole-slide images, have advanced predictions of response to NAT of breast cancer. Future work will explore incorporating additional imaging modalities, such as pathological slides and ultrasound, which may enhance the performance of our models and contribute to their overall improvement.”

Reviewer #2 (Remarks to the Author): clinical expertise in breast cancer therapy response

In this research, Gao and colleagues embarked on a pioneering project to develop a Multi-modal Response Prediction (MRP) for early-stage breast cancer, leveraging clinical-pathological and radiological imaging data. The endeavor to identify predictors of pathological complete response (pCR) is a pivotal area of interest in breast cancer research. To date, numerous biomarkers have been scrutinized for their predictive value using pre-treatment baseline characteristics across various breast cancer subtypes and therapeutic approaches. The significance of pCR status is well-established in HER2-positive and triple-negative breast cancer due to its profound correlation with enhanced long-term survival. However, its relevance in HR+/HER2- breast cancer remains ambiguous. Additionally, the specific therapeutic agents achieving pCR are critical, as they bear implications for long-term outcomes.

We would like to thank the referee for their positive and thorough comments. We are pleased that the referee recognizes the pioneering project we have conducted is a pivotal area of interest in breast cancer research. We agree that the value of pCR for long-term outcome prediction is more established for HER2-positive and triple-negative breast cancer subtypes than for ER+ disease. However, patients with ER+/HER2- disease who achieve a pCR still have a more favorable prognosis compared to patients with no pCR (Yau et al. Lancet Oncol. 2022) ⁷. In addition, accurate prediction of pCR may have profound implications for therapy such as the extensiveness of surgery (if any), and enrolment into neoadjuvant clinical trials of novel therapies.

While the authors have undoubtedly invested a considerable effort in this study, meriting recognition, I harbor significant reservations regarding the originality of their findings and their impact on the breast cancer domain. My concerns are as follows:

The bulk of the predictive outcomes and their comparisons with interpretations by radiologists, as well as with other clinical-pathological indicators, are derived from a substantial training dataset.

We acknowledge with the reviewer that the model was validated on an in-house test set (cohort A, which is independent of training and validation sets) and an external public dataset from Duke University (cohort B). Both were evaluated along with human reader study, as illustrated in Fig. 3 for cohort A and Supplementary Fig. 4b for cohort B.

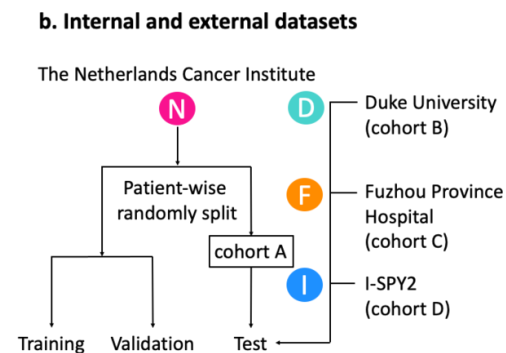
This comment is related to comment 2 below. We assume the reviewer's concern is that the external datasets with extensive data would provide for a more comprehensive validation of the model efficiency. Therefore, we have collected two additional external data sources involving breast cancer patients who received neoadjuvant therapy (NAT): one from a private center (cohort C, n=85) and one from public center (cohort D, n=508). Cohort C comprises pre-NAT mammograms and longitudinal MRI data before

and after NAT; cohort D includes longitudinal MRI data pre-, mid- and post-NAT. Both cohorts include longitudinal multi-modality data, allowing a more comprehensive validation of the proposed models comparisons, including iMGrhpc, iMRrhpc, and MRP.

Action taken: This comment is related to comment 2 below, and we have addressed both comments by:

(1) We have added cohort diagrams to Fig. 2 and sample collection details to Supplementary Fig. 1, with case information in Supplementary Tabs. 7 and 8 (page 16, lines 484-496);

Lines 484-496 (Cohorts introduction):



New Figure 2b: Internal and external cohorts

“Fujian Provincial Hospital Dataset (cohort C). The cohort consists of 220 breast cancer patients who received neoadjuvant therapy between January 1st, 2019, and December 31st, 2023, collected from Fujian Provincial Hospital (FJPH) in China. The dataset comprises pre-NAT mammograms, longitudinal dynamic contrast-enhanced MRI data before and after NAT, clinical information, and histopathologic outcome data. To ensure comparable validation between the iMGrhpc and iMRrhpc models across the NAT period, we excluded 135 patients due to either missing multi-modal imaging or longitudinal MRI, or unqualified imaging, as shown in Extended Data Fig. 1. (c). The remaining 85 cases were included. For further patient details, please refer to Extended Data Tab. 7.

I-SPY2 Dataset (cohort D). In the I-SPY2 (Investigation of Serial studies to Predict Your Therapeutic Response with Imaging And moLecular analysis 2) ^{8,9} collection, from 2010 to 2016, 719 breast cancer patients were enrolled across multiple clinical sites in the United States, including dynamic contrast-enhanced MRI and clinicopathologic data. In this study, we excluded 211 patients who were either size mismatched between pre- and postcontrast sequences or missing multi-time points (pre-, mid-, and post-NAT) MRIs, as shown in Extended Data Fig. 1. (d). Ultimately, 508 cases and corresponding 2,032 imaging exams were included in the I-SPY2 dataset. For further patient details, please refer to Extended Data Tab. 8.”

(2) We have included the model's comparison results in Supplementary Figs. 6-9 and Tab. 2 (page 5, lines 129-143). The findings from these cohorts confirmed the efficiency of the developed multi-modal system for predicting the response to neoadjuvant therapy in breast cancer.

Lines 129-143 (Model's performance description):

"In cohort C, as illustrated in Extended Data Fig. 6, the model's performance consistently improves with the addition of successive levels of multi-modal imaging and longitudinal MRIs. At the pre-NAT stage, compared to the uni-modal rhpc model, MRP increases the AUROC to 0.82, representing an 8.4% improvement over the rhpc model, and AUPRC to 0.73, reflecting a 17.2% increase. At the post-NAT stage for the longitudinal scenarios setting, MRP achieves an AUROC of 0.84 and AUPRC of 0.76. The differences in AUC between the rhpc and MRP models are statistically significant at each NAT stage (pre-NAT: $p = 0.009$, post-NAT: $p = 0.006$). For detailed results on breast cancer molecular subtypes, please refer to Extended Data Fig. 7.

In cohort D, as illustrated in Extended Data Fig. 8, the model achieves superior performance with the integration of multi-modal data and longitudinal MRIs. The iMRrhpc model, using pre-NAT data, shows a 12.3% increase in AUROC (0.80) and a 16.3% increase in AUPRC (0.70) compared to the uni-modal rhpc model. Further integration of pre- and mid-NAT data results in AUROC improvements of up to 15% (0.83) and AUPRC improvements of up to 20.3% (0.74). Additionally, pre- and post-NAT data integration leads to AUPRC improvements of up to 20.8% (0.75). Compared to the rhpc model, the multi-modal model that integrated imaging demonstrates a significant performance improvement at each NAT scenario ($p < 0.001$). For details on the results of breast cancer molecular subtypes, please see Extended Data Fig. 9."

The performance metrics within the genuinely independent dataset (i.e., the Duke cohort) leave much to be desired. Furthermore, the sample size of this independent dataset is notably small, particularly for certain subsets such as HER2+ or those receiving endocrine therapy, complicating the reliability of any derived conclusions.

Performance (the Duke cohort): In the Duke dataset (cohort B), mammograms and longitudinal MRIs data are not available, and important clinicopathologic variables such as tumor histology type, in situ component findings and tumor differentiation degree are missing. These factors, proven predictors of breast cancer neoadjuvant therapy response in previous studies^{10–13}, significantly contributed to the overall training model, as depicted in Fig.4. As a consequence, the model's performance in the Duke cohort is inferior compared to scenarios where all information were available.

In addition, the results on the FJPH (cohort C) and the I-SPY2 (cohort D) datasets demonstrated the model's efficiency in integrating longitudinal multi-modal imaging compared to models utilizing less complete information. For a detailed description of these results, please refer to the last comment. These results highlight the need for increased access to comprehensive clinical information and imaging modalities in public datasets to advance multi-modal AI research.

To mitigate the impact of incomplete clinical data on multi-modal AI system and improve data efficiency, we deliberately designed a cross-modal learning technique that mines clinically relevant features from routinely available imaging data. Ablation studies (Supplementary Figs. 12-14) showed that this technique led to an average 7.6% increase in AUC, with cohorts B, C, and D showing improvements of 8.6%, 6.2%, 8.0%, respectively, compared to the model without cross-modal learning (see discussion on page 11, lines 345-354 of manuscript).

Sample size: After including additional test sets, the total number of patients from external cohorts increased from 288 to 881. In particular, the HER2+ patients increased from 60 to 222 (Supplementary Fig.1). In this way, these external validations have provided a more thorough test of the model efficiency. Additionally, this study represents the biggest breast cancer cohort in the literature we know of that integrates multi-modal breast imaging and clinicopathological data with neoadjuvant therapy response outcomes assessed after surgery.

Action taken: We have discussed the sample size on page 13, lines 393-395: *"Although this study represents the biggest breast cancer cohort in the literature to our knowledge that integrates longitudinal multi-modal breast data with NAT outcomes, further prospective clinical trials are required for enhancing the model's clinical evidence."*

The models used prior to neoadjuvant therapy (pre-NAT) appear to yield results comparable to those of other models that consider imaging during and after treatment. This raises the question of whether these pre-NAT models surpass existing predictive biomarkers like OncotypeDX, HER2DX, or the model recently introduced by Carlos Caldas's team in Nature Cancer Cancer (<https://www.nature.com/articles/s41586-021-04278-5>).

The use of on-treatment MRIs is not uniformly accepted worldwide nor endorsed by clinical guidelines.

OncotypeDX, HER2DX: We do not have genetic test data for these patients, as it's not currently routinely performed in most practices in the Netherlands. Therefore, we cannot directly compare results. However, genetic tests are costly and require subsequent testing. The major expectation of our model is that it provides an accurate prediction using readily available data for virtually any breast cancer patient. In the

practice of our country, MRI scans are reimbursed and much more available, unlike genetic tests. Notably, our released code allows readers to build models using different predictive assays, such as OncotypeDX and HER2DX. Therefore, we followed the reviewer’s other suggestion on comparing our method with the method ⁵ by Carlos Caldas’s team.

Predictive model comparison: As a comparison, we assessed the performance of the multi-omic predictor described in research ⁵ by Carlos Caldas’s team. Although the scope of this predictor is very different from the proposed model in deployed features (for example, Ref. ⁵ mainly relies on RNA-seq and DNA features), both models emphasize the fundamental value of established clinical variables in the neoadjuvant setting. It is noteworthy that the AUCs achieved by their model were slightly lower than those obtained by our clinical model (please see Table R1.1).

Model	Cohort B	Cohort C
Ref⁵ -Clinical		
AUROC	0.65 (0.58-0.71)	0.69 (0.57-0.80)
Ours-Clinical		
AUROC	0.66 (0.60-0.73)	0.73 (0.62-0.83)
iMRrhpc (pre-NAT)		
AUROC	0.80 (0.74-0.85)	0.80 (0.70-0.89)

Table R1.1 Prediction performance of iMRrhpc compared with Ref ⁵ and ours clinical based models using same clinical characteristics as Ref ⁵ on test sets.

Action taken: We have added the following sentences at page 10 lines 288-292 to illustrate this: “*The baseline clinical model, implemented using our framework, demonstrated similar performance to a previously published clinical predictor ⁵ using the same clinical features. In contrast, our NAT routine-used imaging (pre-NAT only) combined model showed a significant improvement over the baseline clinical model (cohort B: iMRrhpc: $p = 2.358e-03$; cohort C: iMRrhpc: $p = 7.584e-03$; MRP: $p = 4.956e-04$).*”

on-treatment MRI: We acknowledge the use of on-treatment MRIs is not uniformly accepted worldwide nor endorsed by clinical guidelines. Given the variability in real-world practices, as also reflected by the shown external data sources, our model was designed to handle the absent MRI input (or other modality data) at different stages, during both training and inference (line 358 and line 361 of manuscript). This also addressed the limitations of previous techniques in handling different NAT practices of hospitals, as suggested by prior studies ^{14,15}.

The authors' conclusion that the MRP could "reduce unnecessary treatment toxicity in non-pCR cases by 35.8% without halting effective treatment in the Pre-NAT scenario, and potentially eliminate surgery for pCR cases by 16.7% without foregoing treatment opportunities in the Post-NAT scenario" is, in my view, flawed. This is because there is no established efficacy for altering therapy based on an anticipated lack of pCR. Moreover, forgoing surgery is not a current standard of care, and the necessity for prospective trials (some of which are currently underway) renders this conclusion somewhat misleading.

We were pleased that the reviewer recognizes MRP's performance. We understand the reviewer's concern and agree that the hypothetical advantages listed here do require substantial further testing/clinical trial in clinical practice. Therefore, predictive power of our algorithm may be used as a basis for further clinical trials in the mentioned directions:

1. In the pre- and mid-NAT setting, identifying patients with treatment-resistant tumors early is crucial, as these patients are unlikely to respond to standard neoadjuvant therapies. They typically have a poorer prognosis and lower rates of breast-conserving surgery¹⁶. Standard of care in these non-pCR patients would consist of completing the full NAT, surgery and then post-NAT systemic treatment (i.e., capecitabine and/or olaparib for triple-negative subtype and TDM-1 for HER2 positive subtype). Identifying these patients earlier would enable enrollment in clinical trials^{16,17} with alternative drug combinations, as standard therapies may not be optimal for them.
2. In the post-NAT setting, patients with treatment-sensitive tumours who will have responded well to standard neoadjuvant therapies may be suitable to more conservative management, following the promising outcomes from feasibility studies (Kuerer et al. Lancet Oncol. 2022, Johnson et al. J Am Coll Surg. 2023)^{18,19}, in which researchers investigated the potential of omitting surgery after neoadjuvant systemic therapy. The developed model may therefore be used to identify these patients, who are most likely to benefit from further clinical trials.

Action taken: Considering the suggestion from the reviewer, as well as the breast oncologists and surgeons in the team, we have moderated our statements taking into account published studies and underway trials on reducing surgery. We have amended the text on page 13 lines 408-412 and as well as amended Figure 2d.

"In the clinical workflow, the MRP could be applied to patients across neoadjuvant therapy stages. In the pre- and mid-NAT setting, patients predicted to have treatment-resistant tumours could be promptly considered for enrolment into neoadjuvant clinical trials of alternative therapies^{16,17}. In the post-NAT setting, patients predicted to have achieved pCR may be considered for clinical trials of reducing surgery^{18,19}, contingent upon further validation."

Given these points, while the study represents a significant undertaking, its contributions to the field of breast cancer, particularly in terms of novelty and clinical applicability, require a more critical examination.

We were pleased to see that referee recognizes our work. We believe this study provides significant steps forward in the highly interdisciplinary breast cancer field. We would like to highlight the following novelties:

(1) From a model design of view, MRP is the first model that allows flexible implementation at any NAT stage, whilst being compatible with missing modalities data. The designed **cross-modal knowledge learning** technique mitigates impact of incomplete data (Supplementary Figs. 12-14, lines 253-254 of the manuscript). This design leveraged imaging feature to extract the clinically relevant information, enhancing efficiency and contribution of routinely used imaging to the prediction model (Fig. 4 and Supplemental Fig. 18, lines 260-261). Additionally, the **temporal information embedding** strategy enables the model to understand the longitudinal imaging information (i.e. Supplementary Figs. 16 and 17, lines 265-266), which further increases input adaptability. These techniques, which enhance the model's potential application in diverse NAT settings in hospitals, are indeed novel, as noted by Reviewer #1.

(2) To our best knowledge, we are also the first to show and quantify, the potential benefits and contributions of integrating histopathological, clinical, personal, and radiological imaging information throughout the NAT patient journey to enhance the performance of predicting pathological complete response in breast cancer.

(3) We are first to show how the comparable robustness and generalizability between international breast radiologists team and AI models (along with error analysis) is associated with the degree of tumor therapy response.

For the clinical applicability:

(1) In order to examine its applicability in different centers and different longitudinal clinical scenarios, we have evaluated our model in different external validation datasets (2 US and 1 Asian dataset). For the validation results descriptions, as seen in comment 1.

(2) In terms of multi-modal imaging data for our model applicability, the most immediate translational potential lies in imaging modalities used in our model that are virtually always available in NAT practice without extra costs. Breast MRI is currently the preferred imaging modality for assessing tumor response to NAT²⁰. Mammography remains the standard imaging technique for breast cancer screening and diagnosis in many clinical settings²¹ and can be used as an independent predictor or integrated to the multi-modal prediction.

(3) In light of the published studies and ongoing clinical trials, we agree with the reviewer that the developed model could, on one hand, be used to select non-pCR patients in the pre-/mid-NAT phase who may benefit the most from the prompt enrollment in alternative therapy trials^{16,17}. On the other hand, in the post-NAT phase, the model could serve as a basis for identifying pCR patients who are most likely to benefit from surgery reducing trials^{18,19}, contingent upon further validation.

(4) Further, the complete source code of our model is publicly accessible. The flexibility of our model allows any data modality to be added or removed to generate predictions to handle clinicopathologic/imaging data variations. This multi-modal data integration pipeline can also be applicable to other tumor types, depending on the data available in any specific study.

Reviewer #3 (Remarks to the Author): expert in breast cancer radiology

NCOMMS-24-02259

Summary

In this retrospective study, the authors develop and validate an explainable longitudinal multi-modal fusion model, the Multi-modal Response Prediction (MRP) system, to predict NAT response in women with breast cancer. The model was developed and tested using data from 3,352 patients with breast cancer who were treated with NAT in the Netherlands and externally tested on a dataset from Duke University. The MRP is composed of two independently trained models: one that includes pre-NAT mammographic imaging and radiological, histopathological, personal, and clinical variables and one that includes longitudinal MR imaging and the aforementioned variables. The MRP system achieved an AUROC of 0.882 for pCR prediction prior to the initiation of NAT and an AUROC of 0.889 for having achieved pCR during and after NAT. The MRP system also outperformed radiologist readers in pCR prediction prior to the initiation of NAT.

Strengths

- A. This topic is of high clinical relevance and importance – if NAT response can be predicted with accuracy, then NAT regimens could be modified early on based on patient response and potential toxicities.
- B. The MRP system integrates multi-modal data from a large cohort of breast cancer patients who underwent NAT (n=3,352).
- C. The authors also conducted a reader study and an error analysis to better understand discrepancies between MRP and radiologist predictions.
- D. The tables and figures are detailed and comprehensive.

[We thank the referee for their very positive comments, and for recognizing the clinical importance of this translational study.](#)

Weaknesses

1. The MRP system was validated on an external dataset from Duke University; however, that external dataset included only pre-NAT MRI examinations (no other MRI examinations or pre-NAT mammographic images), and thus the MRP system was not comprehensively tested.

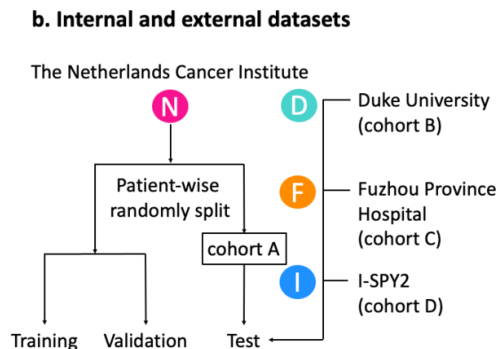
We acknowledge that the MRP system was initially validated on an external public dataset from Duke University that includes only pre-NAT MRI examinations. It is a well-built, large and publicly available dataset to validate our findings. We understand that external datasets with pre-NAT mammograms and longitudinal MRIs would provide for a more comprehensive validation of the MRP system.

Validation dataset: We have invested our efforts in collecting two additional external datasets, cohort C and D, with 85 and 508 samples, respectively, from breast cancer patients who received NAT. Cohort C comprises pre-NAT mammograms and longitudinal dynamic contrast-enhanced MRI data before and after NAT. Cohort D includes longitudinal MRI data pre-, mid- and post-NAT. Both cohorts include longitudinal multi-modal data, allowing a more comprehensive validation of the proposed models including iMGrhpc, iMRhpc, and MRP.

Action taken: We have addressed this comment by:

- (1) We have added cohort diagrams to Fig. 2 and sample collection details to Supplementary Fig. 1, with case information in Supplementary Tabs. 7 and 8 (page 16, lines 484-496);

Lines 484-496 (Cohorts introduction):



New Figure 2b: Internal and external cohorts

“Fujian Provincial Hospital Dataset (cohort C). The cohort consists of 220 breast cancer patients who received neoadjuvant therapy between January 1st, 2019, and December 31st, 2023, collected from Fujian Provincial Hospital (FJPH) in China. The dataset comprises pre-NAT mammograms, longitudinal dynamic contrast-enhanced MRI data before and after NAT, clinical information, and histopathologic outcome data. To ensure comparable validation between the iMGrhpc and iMRhpc models across the NAT period, we excluded 135 patients due to either missing multi-modal imaging or longitudinal MRI, or unqualified imaging, as shown in Extended Data Fig. 1. (c). The remaining 85 cases were included. For further patient details, please refer to Extended Data Tab. 7.

I-SPY2 Dataset (cohort D). In the I-SPY2 (Investigation of Serial studies to Predict Your Therapeutic Response with Imaging And moLecular analysis 2) ^{8,9} collection, from 2010 to 2016, 719 breast cancer patients were enrolled across multiple clinical sites in the United States, including dynamic contrast-enhanced MRI and clinicopathologic data. In this study, we excluded 211 patients who were either size mismatched between pre- and postcontrast sequences or missing multi-time points (pre-, mid-, and post-NAT) MRIs, as shown in Extended Data Fig. 1. (d). Ultimately, 508 cases and corresponding 2,032 imaging exams were included in the I-SPY2 dataset. For further patient details, please refer to Extended Data Tab. 8.”

(2) We have included the model’s comparison results in Supplementary Figs. 6-9 and Tab. 2 (page 5, lines 129-143). The findings from these cohorts confirmed the efficiency of the developed multi-modal system for predicting the response to neoadjuvant therapy in breast cancer.

Lines 129-143 (Model’s performance description):

“In cohort C, as illustrated in Extended Data Fig. 6, the model’s performance consistently improves with the addition of successive levels of multi-modal imaging and longitudinal MRIs. At the pre-NAT stage, compared to the uni-modal rhpc model, MRP increases the AUROC to 0.82, representing an 8.4% improvement over the rhpc model, and AUPRC to 0.73, reflecting a 17.2% increase. At the post-NAT stage for the longitudinal scenarios setting, MRP achieves an AUROC of 0.84 and AUPRC of 0.76. The differences in AUC between the rhpc and MRP models are statistically significant at each NAT stage (pre-NAT: $p = 0.009$, post-NAT: $p = 0.006$). For detailed results on breast cancer molecular subtypes, please refer to Extended Data Fig. 7.

In cohort D, as illustrated in Extended Data Fig. 8, the model achieves superior performance with the integration of multi-modal data and longitudinal MRIs. The iMRrhpc model, using pre-NAT data, shows a 12.3% increase in AUROC (0.80) and a 16.3% increase in AUPRC (0.70) compared to the uni-modal rhpc model. Further integration of pre- and mid-NAT data results in AUROC improvements of up to 15% (0.83) and AUPRC improvements of up to 20.3% (0.74). Additionally, pre- and post-NAT data integration leads to AUPRC improvements of up to 20.8% (0.75). Compared to the rhpc model, the multi-modal model that integrated imaging demonstrates a significant performance improvement at each NAT scenario ($p < 0.001$). For details on the results of breast cancer molecular subtypes, please see Extended Data Fig. 9.”

2. Clarification about the ground truth reference standard should be provided – in particular, the “unified pCR standard” should be defined and whether the presence of DCIS (but not invasive disease) is considered pCR.

We agree that clarification regarding the ground truth reference standard is essential. We defined pCR as the absence of invasive disease in breast and axillary nodes (ypT0/is, ypN0), which is the most commonly used definition in the literature ²² and allows for the presence of residual DCIS (ypTis) in the pathological specimen.

Action taken: We have detailed the pCR definition standard on page 16, lines 498-501: “pCR is defined as the absence of invasive disease in breast and axillary nodes (ypT0/is, ypN0) at time of surgery, which allows for the presence of residual DCIS (ypTis) in the pathological specimen.” and lines 501 to 502: “Note that for multi-center cohorts, a unified pCR standard ²² was used.”

3. The authors should clarify how the 120 patients for test cohort A were selected. Since all had mammogram and MRI examinations available for analysis, their selection does not seem random. Furthermore, the authors should verify that all reported AUROC values refer to test set performance.

Test cohort A: We randomly selected 120 patients for test cohort A. We first divided the MG dataset, consisting of 1,945 patients, randomly allocating 195 patients (10% of the total) to the test set. Out of these 195 patients, 120 had images available at both the pre-NAT mammogram and at three different MRI examination time points (pre-, mid-, and post-NAT), which allows for model evaluation across these stages in a multi-modal fashion. The remaining patients ($195 - 120 = 75$) were added to the validation set ($194 + 75 = 269$), ensuring that the validation and test sets comprised 20% of the total. Consequently, for the MRI dataset division, we maintained a consistent test set of 120 patients with the MG dataset for comparative purposes, while still preserving a 20% set for validation and test. We have clarified this in the revision.

Reported performance values: We have added the sentence to verify that all reported AUROC values refer to test set performance on page 5, line 123.

Action taken: This comment is related to comment 13 below, and we have addressed both comments by detailing the dataset randomization on page 15, lines 462-468.

“In total, 3,352 patients were included in this multi-modality study, where pre-NAT MG and multi-timepoint MRI datasets were collected. We first randomly divided the MG dataset at the patient level, allocating 80% for training and the remaining 20% split equally for validation and testing. In the test subset, 120 patients

were further selected to form the test cohort A. These patients had both available pre-NAT MG and longitudinal MRI examinations at three different time points (pre-, mid-, and post-NAT), allowing for model evaluation across these stages. The remaining cases from the test subset were added to the validation set. Consequently, for the MRI dataset division, we ensured a consistent test set of 120 patients with the MG dataset for comparative purposes, while still maintaining a 20% allocation for validation and test. “

4. Ultrasound and MRI are most commonly used to monitor NAT response. The authors should address in the Discussion section why mammography rather than ultrasound was used in the MRP system.

We agree that ultrasound (US) has demonstrated utility in predicting pathologic tumor size ²⁰ and assessing response in axillary lymph nodes ²³ following neoadjuvant therapy. However, US is known to have a high interobserver discordance and also limited reproducibility due to its operator-dependency ²³. The most accurate images may not always be stored, which implies that it may be a potential confusion factor for the model. On the other hand, mammography has significantly higher reproducibility than US ^{24,25} and has been part in other types of computer-aided detection systems ²⁶. Even though mammography is not usually used to monitor the tumor response during the treatment, it remains the standard imaging technique for breast cancer screening and diagnosis in many clinical settings ²¹. It is virtually always available for primary tumor staging and thus can still be used as an independent predicting factor in the pre-NAT scenario or integrated to the multi-modal prediction.

While we have chosen not to include ultrasound in this study due to its operator dependency, we aim to work towards incorporating it into our model in the future, considering its effectiveness in predicting tumor size ²⁰ and assessing lymph node status ²³. Although this will require substantial data curation, retraining the deep learning architecture described in this manuscript and further validation, the benefit of our model is that features from any data modality can be added or removed to generate predictions.

Action taken: We have addressed this comment in discussion section (page 10, lines 317-324), as follows.

“Regarding the applicability of multi-modal imaging, breast MRI is currently the preferred imaging modality for assessing tumor response to neoadjuvant therapy ²⁰. Mammography remains the standard imaging technique for breast cancer diagnosis ²¹ and can be used as an independent predictor in the pre-NAT scenario. While we have chosen not to include ultrasound in this study due to its operator dependency ²³, we aim to work towards incorporating it into our model in the future, considering its effectiveness in predicting tumor size ²⁰ and assessing lymph node status ²³. Although this will require

retraining the deep learning architecture described in this study and further validation, the benefit of our model is that features from any data modality can be added or removed to generate predictions. "

5. More detail and discussion about the modality contribution analysis would be helpful (specific questions below). Figure 5 suggests that radiological features are less important than histopathological and other features.

Indeed, as the reviewer rightfully notes, the contribution of histopathological features to the overall training model is greater than that of radiological features. By including these features, we have shown that the biology of the tumor predominantly drives the prediction of response to therapy. This has been previously confirmed clinically by others ^{5,27}, where tumor expression was a stronger predictor of response. In this study, histopathological data comprises tumor-specific features from tumor histology type and tumor-cell proliferation, as well as their functional states, whilst the radiological data predominantly captures tumor-phenotype features. Nevertheless, as there was tumor response heterogeneity in the breast density ^{28,29} and tumor distribution (unifocal or multifocal) ³⁰, it is important to include these radiological variables to explore whether a gain could be obtained from adding this. Original Fig. 5 (now referred to as Fig. 4) also confirms that there is some benefit to include images in the three models.

Action taken: More details about the modality contribution analysis are addressed in comment 19 below. We have added the modality contribution discussion on page 10, lines 304-307.

"Histopathological features had the largest contribution to the prediction of response (Fig. 4), with tumor-specific features like histology type and molecular subtype being the most important. Although radiological data, which capture tumor-phenotype features, contributed less overall, the heterogeneity of breast cancer response related to mammographic density ^{28,29} and tumor distribution (unifocal or multifocal) ³⁰ demonstrated additional predictive power in our model."

6. In the Methods section, the authors should clarify what criteria are used at their institution to determine whether or not a patient undergoing NAT is monitored with MRI.

At our institution, all patients undergo breast MRI prior to and after NAT. Between 2010 and 2015, an interim scan was standard. Before 2010, and after 2015, this practice was performed for patients enrolled in trials or when the effectiveness of the treatment was uncertain. Additionally, acknowledging this variability in real-world NAT practices, our model is designed to be flexible; it can integrate or exclude

features from any data modality/timepoint to adapt predictions during both training and inference phases (refer to lines 358 and 361).

Action taken: We have provided clarification on the criteria of MRI used for monitoring patients undergoing NAT on page 15, lines 452-454.

“All patients treated at our institution undergo pre- and post-NAT breast MRI. Interim scans were standard between 2010 and 2015. Before 2010, and after 2015, this practice was performed for trial participants or when treatment efficacy was in doubt.”

Specific Comments

7. Title: Please consider “...for predicting neoadjuvant therapy response in women with breast cancer.”

We agree that the reviewer’s advice for title and it is changed to “An explainable longitudinal multi-modal fusion model for predicting neoadjuvant therapy response in women with breast cancer”.

8. Abstract: Please consider changing “clinical reality” to “the real-world clinical setting.”

We thank the reviewer for their remark and revised relevant texts in the abstract.

9. Abstract: Please consider changing “real clinical pathway” to “pathways used in real-world clinical settings.”

We have revised relevant texts accordingly.

10. Abstract: Please spell out pCR upon first use.

We have amended the manuscript and ensured that abbreviations (including cTNM, CI) are defined the first time they are used.

11. Introduction, line 27: Please clarify what is meant by “radical” surgery.

We agree that the term “radical surgery” may have caused confusion, as we intended to describe the surgery as completely curative. We have removed this term from the revised manuscript.

12. Introduction, line 46: Please correct “patient prognosis care.”

We have corrected the sentence, please see below.

“Additionally, integrating histopathological information has also been shown to complement tumor heterogeneity assessment and enhance the predictive capacity of DL models for patient prognosis^{5,31–33}”

13. Results, Patient Data: How were the 120 patients for test cohort A selected? Since all had mammogram and MRI examinations available for analysis, their selection does not seem random.

Action taken: The 120 patients were randomized into the test cohort A. We have addressed this point in comment 3, more details on the randomization are provided. See page 15 lines 462-468.

“In total, 3,352 patients were included in this multi-modality study, where pre-NAT MG and multi-timepoint MRI datasets were collected. We first randomly divided the MG dataset at the patient level, allocating 80% for training and the remaining 20% split equally for validation and testing. In the test subset, 120 patients were further selected to form the test cohort A. These patients had both available pre-NAT MG and longitudinal MRI examinations at three different time points (pre-, mid-, and post-NAT), allowing for model evaluation across these stages. The remaining cases from the test subset were added to the validation set. Consequently, for the MRI dataset division, we ensured a consistent test set of 120 patients with the MG dataset for comparative purposes, while still maintaining a 20% allocation for validation and test.”

14. Results, Primary Results, line 100: Please clarify what is meant by “During the Pre-NAT period” – does this refer to pre-NAT mammographic and MR imaging?

We agree with the reviewer that “During the Pre-NAT period” is less appropriate to state the time phase here. And as reviewer understands, in the pre-NAT phase, the MRP model utilized pre-NAT mammographic and MR images for inferencing.

Action taken: We have amended the phrase and the respective figure specified (line 106) and ensured that these kinds of descriptions (line 112, 117) are clarified.

15. Results, Primary Results, line 103: Please clarify what is meant by the uni-modal rhpc model.

We have modified the sentence to *“When compared to the uni-modal (i.e., without radiological images integrated) rhpc model, ...”*

16. Results, Primary Results, lines 105-106: Given that the improvement is not statistically significant, please consider tempering the statement that “the MRP’s performance surpasses that of the Pre-NAT period.”

We have rephrased the sentence as *“the MRP’s performance moderately enhances that of the Pre-NAT period”*.

17. Results, Primary Results, line 122: The paragraph beginning with “MRP outperforms six international breast radiologists...” is difficult to follow.

We have revised the paragraph beginning with *“MRP demonstrates favorable performance to breast radiologists in pCR prediction on both the in-house and external datasets.”*

18. Results, Modality Contribution Analysis, lines 149-150: What is meant by in situ status (invasive + in situ disease)? What is meant by density in this context (mammographic breast density)?

Yes, the reviewer is correct. “In situ status” refers to the concurrent presence of an in situ component next to the invasive tumor based upon the biopsy results. “Density” in this context indeed refers to mammographic breast density.

Action taken: We have specified the meaning of these variables in the manuscript, please see page 6, line 167.

“in situ status (presence of an in situ component next to the invasive component)”, and line 171 *“mammographic density”*.

19. Results, Modality Contribution Analysis: With regard to the top five significant features, what is the direction of the association with pCR?

We agree that it's important to clearly communicate the direction of top five significant features' association with pCR. This information could in part be obtained from Tab. 1. However, we acknowledge that this could be more clearly presented.

Action taken: We have added the corresponding main text (page 6, lines 176-189), as follows.

"In addition, we evaluated the association between pCR and the top five contributing factors identified in the MRP model. As indicated in Tab.1 and Extended Data Tab.2, the pCR varied by molecular subtype. Compared to ER+/HER2- cancers (3.8% pCR rate, odds ratio (OR): 1, as reference), the highest pCR OR was observed for ER-/HER2+ (54.9%, OR: 9.18, $P = 0.021$) and TNBC (50.1%, OR: 7.02, $P < 0.001$) subtypes. Many studies have reported that in situ components show lower proliferation rates, and their non-invasive nature contributes to chemoresistance^{12,13}. Our study confirmed that a known in situ component is predictive of poorer response to therapy (24.8%, OR: 0.51, $P < 0.001$). Regarding tumor histology characteristics, patients with ductal carcinoma (36.2%, OR: 6.19, $P < 0.001$) have a higher likelihood of achieving pCR compared to those with lobular carcinoma (6.2%, OR: 1.00, as reference). This observation aligns with previous reports^{10,11} and underscores the need for histology-specific treatment approaches in breast cancer management. Consistent with previously published study³⁰, multi-focal tumors (24.1%, OR: 0.53, $P < 0.001$) exhibited a lower rate of pCR compared to uni-focal tumors (41%, OR: 1.00, as reference). Prior studies^{28,29} suggest considering mammographic density in planning and optimizing treatment strategies for breast cancer patients. However, while our study demonstrates a slightly higher likelihood of pCR in patients with dense breasts (20.5%, OR: 1.05, $P = 0.159$) compared to patients with less dense breasts (18.8%, OR: 1.00, as reference), this was not statistically significant."

20. Results, Personalized Management: Please correct subtitle "bilateral caring."

We have corrected the subtitle to "*Personalized management considering both non-pCR and pCR patients*".

21. Results, Personalized Management, line 194: Please clarify what is meant by "although radiologists offered a better choice than the standard clinical approach..."

We have clarified this sentence in the revised manuscript (page 8, lines 222-224).

“When comparing our model with human readers, although there is a benefit offered by adding radiologists findings to the standard clinical approach (black curve, i.e., $y=0$), the MRP (red curve) still outperformed human readers at each time point.”

22. Discussion: Ultrasound and MRI are most commonly used to monitor NAT response. Please address why mammography rather than ultrasound was used in the MRP system.

Action taken: We have addressed this point in comment 4. We added the related discussion on page 10, lines 317-324, as follows.

“Regarding the applicability of multi-modal imaging, breast MRI is currently the preferred imaging modality for assessing tumor response to neoadjuvant therapy²⁰. Mammography remains the standard imaging technique for breast cancer diagnosis²¹ and can be used as an independent predictor in the pre-NAT scenario. While we have chosen not to include ultrasound in this study due to its operator dependency²³, we aim to work towards incorporating it into our model in the future, considering its effectiveness in predicting tumor size²⁰ and assessing lymph node status²³. Although this will require retraining the deep learning architecture described in this study and further validation, the benefit of our model is that features from any data modality can be added or removed to generate predictions.”

23. Discussion: In my opinion, the primary benefit of the MRP system is its potential use to modify and personalize NAT regimens. The other proposed use – to avoid definitive surgery – could result in adverse patient outcomes if complete pCR is incorrectly predicted by the system.

We are pleased to see that the referee acknowledges our proposed system’s potential impact in personalizing NAT regimens. Indeed, we agree that while MRP systems can provide valuable insights into treatment response, there is a risk of incorrect predictions, as highlighted in our error analysis, which may lead to adverse outcomes if surgery is deferred based solely on the model. We acknowledge that it's essential for clinicians to interpret MRP's prediction cautiously, using them as a concurrent or second-read tool contingent upon further validation, and to consider other clinical factors and patient preferences when making treatment decisions to avoid the possibility of undersurgery. In the revision, we indicated that, in any case, when AI predictions are used to avoid surgery, oncological safety should be evaluated in a randomized controlled trial.

Action taken: We have added the following discussion on page 13, lines 388-390: *“Further, careful interpretation and integration of the MRP model into clinical practice by clinicians are necessary to ensure*

its safe and effective use, considering multiple factors such as clinical judgment and patient preferences. Safety of adapting regular clinical processes should be tested in randomized clinical trials.”

24. Methods: At the authors' institution, what criteria are used to determine whether or not a patient undergoing NAT is monitored with MRI? Or, is MRI routinely obtained in all patients undergoing NAT?

Action taken: We have addressed this point in comment 6 and provided clarification on the criteria of MRI used for monitoring patients undergoing NAT on page 15, lines 452-454.

“All patients treated at our institution undergo pre- and post-NAT breast MRI. Interim scans were standard between 2010 and 2015. Before 2010, and after 2015, this practice was performed for trial participants or when treatment efficacy was in doubt.”

25. Methods, Inclusion and Ethics Statement, line 394: Relatively few patients (n=120) had both pre-NAT MG and MRI.

In total, 916 patients who had both mammogram and MRI examinations were available in the in-house dataset (train/valid/test distribution: 687/109/120 patients). We randomly selected 120 patients to form the test cohort A (for a detailed description of the randomization procedure, please refer to comment 3). As addressed in comment 1, we collected additional external cohorts. Therefore, the external test set now include 205 patients who had both pre-NAT mammograms and longitudinal MRIs. We hope the reviewer will appreciate the difficulty in obtaining patient samples with extensive multi-modality data. This study represents the biggest breast cancer cohort in the literature we know of that integrates multi-modal breast imaging and clinicopathological data with neoadjuvant therapy response outcomes assessed after surgery.

It is also important to highlight that, similar to the data split training strategy used in a previous study (Nature, 2022) ⁵, which used multi-omic data to predict neoadjuvant responses in breast cancer—with 147 of 168 patients in the training set having multi-omic data, and 75 patients in the test set having multi-omic data—this setup allows each modality-based model to fully utilize its data for effective training. In our study, MG-based and MRI-based models (i.e., iMGrhpc and iMRrhpc) support independent training and inference, designed to fully use the available data and handle the frequently unpaired multi-modal data.

26. Methods, Ground Truth, line 413: Please clarify what is meant by a “unified pCR standard” or provide a reference.

We have provided a reference for the unified pCR standard ²² in the Methods section, on page 16, line 501.

27. Figure 1: Please define “TN.”

We have specified its’ definition in the Figure 1: “..., assessing the post-NAT pathological tumor (ypT) and lymph node staging (ypN) scores (together ypTN) ³⁴, which is the international standard for post-neoadjuvant therapy pathology reporting.”

28. Figure 5: These graphs suggest that radiological features are less important than histopathological and other features. Please comment on this in the Discussion section.

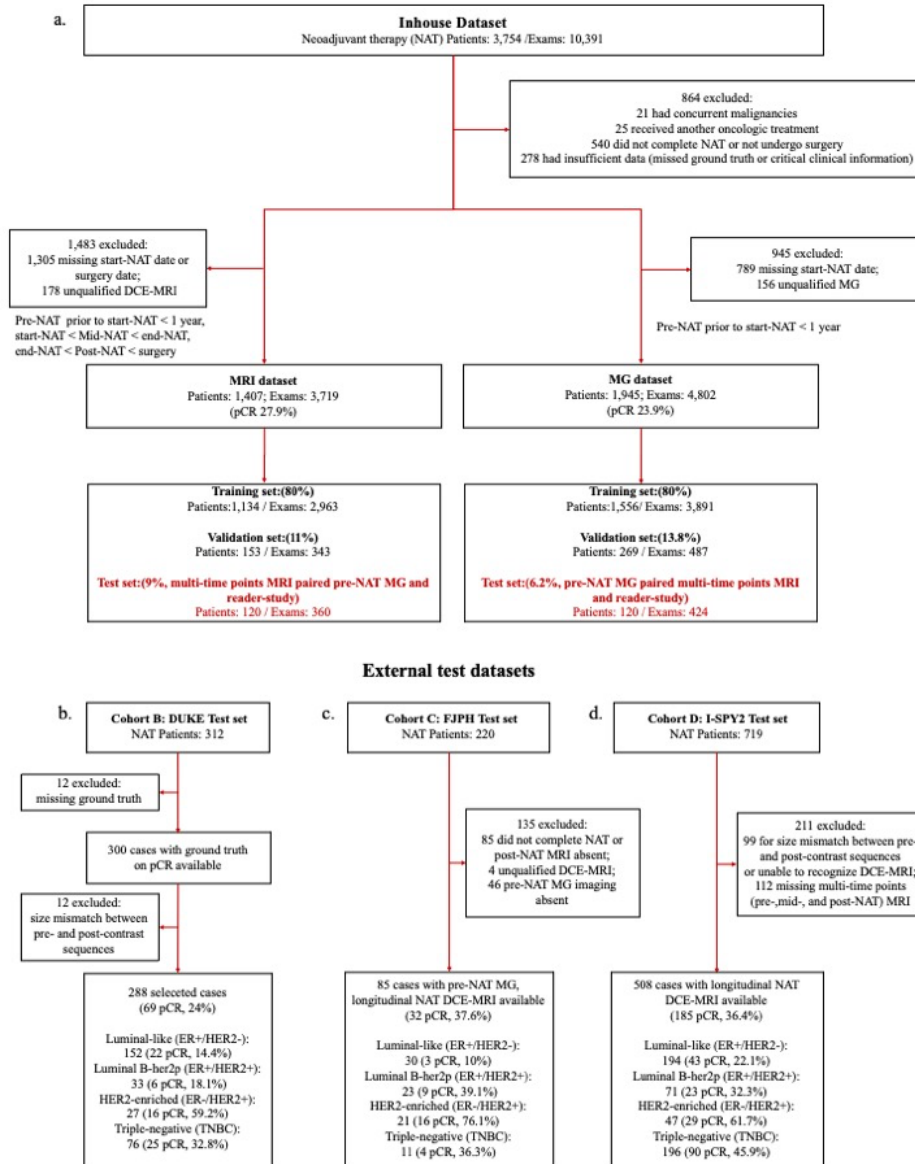
Action taken: We have addressed this point in comment 5 and added the corresponding text in the Discussion section, on page 10, lines 304-307.

“Histopathological features had the largest contribution to the prediction of response (Fig. 4), with tumor-specific features like histology type and molecular subtype being the most important. Although radiological data, which capture tumor-phenotype features, contributed less overall, the heterogeneity of breast cancer response related to mammographic density ^{28,29} and tumor distribution (unifocal or multifocal ³⁰ demonstrated additional predictive power in our model.”

29. Figures: Please consider adding a patient flowchart that details the exclusion criteria.

We agree with the reviewer that it is important to make sure that the patient exclusion process is made clear in the manuscript.

Action taken: We have added the new Supplementary Fig. 1 to include the patient exclusion details.



Supplementary Fig. 1: Amended to address Reviewer 3's comment 29

30. Table 1: To what does location refer (C50.0, etc)? Please clarify.

Action taken: We have included the meanings of the location codes in the footnote of Table 1.

"Breast tumor location is coded according to the International Classification of Diseases for Oncology (ICD-O). C50.0: Nipple; C50.1: Central portion of breast; C50.2: Upper-inner quadrant of breast; C50.3: Lower-inner quadrant of breast; C50.4: Upper-outer quadrant of breast; C50.5: Lower-outer quadrant of breast; C50.6: Axillary tail of breast; C50.8: Overlapping lesion of breast; C50.9: multi-focal neoplasm in more than one quadrant of the breast."

Reviewer #3 (Remarks on code availability):

I was able to access the algorithm code at the following link: <https://github.com/yawwG/MRP/>.

We thank the referee for confirming access to the algorithm code.

References

1. Clevert, D.-A., Unterthiner, T. & Hochreiter, S. Fast and Accurate Deep Network Learning by Exponential Linear Units (ELUs). Preprint at <http://arxiv.org/abs/1511.07289> (2016).
2. Bera, K., Braman, N., Gupta, A., Velcheti, V. & Madabhushi, A. Predicting cancer outcomes with radiomics and artificial intelligence in radiology. *Nat Rev Clin Oncol* **19**, 132–146 (2022).
3. Gorishniy, Y., Rubachev, I. & Babenko, A. On Embeddings for Numerical Features in Tabular Deep Learning. Preprint at <http://arxiv.org/abs/2203.05556> (2023).
4. Ching, T. *et al.* Opportunities and obstacles for deep learning in biology and medicine. *J. R. Soc. Interface*. **15**, 20170387 (2018).
5. Sammut, S.-J. *et al.* Multi-omic machine learning predictor of breast cancer therapy response. *Nature* **601**, 623–629 (2022).
6. Ogier Du Terrail, J. *et al.* Federated learning for predicting histological response to neoadjuvant chemotherapy in triple-negative breast cancer. *Nat Med* **29**, 135–146 (2023).
7. Yau, C. *et al.* Residual cancer burden after neoadjuvant chemotherapy and long-term survival outcomes in breast cancer: a multicentre pooled analysis of 5161 patients. *The Lancet Oncology* **23**, 149–160 (2022).
8. Rugo, H. S. *et al.* Adaptive Randomization of Veliparib–Carboplatin Treatment in Breast Cancer. *N Engl J Med* **375**, 23–34 (2016).
9. Park, J. W. *et al.* Adaptive Randomization of Neratinib in Early Breast Cancer. *N Engl J Med* **375**, 11–22 (2016).
10. Petrelli, F. & Barni, S. Response to neoadjuvant chemotherapy in ductal compared to lobular carcinoma of the breast: a meta-analysis of published trials including 1,764 lobular breast cancer. *Breast Cancer Res Treat* **142**, 227–235 (2013).
11. Tubiana-Hulin, M. *et al.* Response to neoadjuvant chemotherapy in lobular and ductal breast carcinomas: a retrospective study on 860 patients from one institution. *Annals of Oncology* **17**, 1228–1233 (2006).
12. Ernster, V. L., Barclay, J., Kerlikowske, K., Grady, D. & Henderson, C. Incidence of and treatment for ductal carcinoma in situ of the breast. *JAMA* **275**, 913–918 (1996).
13. Burstein, H. J. & Lester, S. C. Ductal Carcinoma in Situ of the Breast. *The New England Journal of Medicine* (2004).
14. Liu, Y. *et al.* Early prediction of treatment response to neoadjuvant chemotherapy based on longitudinal ultrasound images of HER2-positive breast cancer patients by Siamese multi-task network: A multicentre, retrospective cohort study. *eClinicalMedicine* **52**, 101562 (2022).
15. Huang, Y. *et al.* Longitudinal MRI-based fusion novel model predicts pathological complete response in breast cancer treated with neoadjuvant chemotherapy: a multicenter, retrospective study. *eClinicalMedicine* **58**, 101899 (2023).
16. Symmans, W. F. *et al.* Long-Term Prognostic Risk After Neoadjuvant Chemotherapy Associated With Residual Cancer Burden and Breast Cancer Subtype. *JCO* **35**, 1049–1060 (2017).

17. Asselain, B. *et al.* Long-term outcomes for neoadjuvant versus adjuvant chemotherapy in early breast cancer: meta-analysis of individual patient data from ten randomised trials. *The Lancet Oncology* **19**, 27–39 (2018).
18. Kuerer, H. M. *et al.* Eliminating breast surgery for invasive breast cancer in exceptional responders to neoadjuvant systemic therapy: a multicentre, single-arm, phase 2 trial. *The Lancet Oncology* **23**, 1517–1524 (2022).
19. Johnson, H. M. *et al.* Eliminating Breast Surgery for Invasive Cancer with Exceptional Response to Neoadjuvant Systemic Therapy: Prospective Multicenter Clinical Trial Planned Initial Feasibility Endpoint. *Journal of the American College of Surgeons* **237**, 101–108 (2023).
20. Fowler, A. M., Mankoff, D. A. & Joe, B. N. Imaging Neoadjuvant Therapy Response in Breast Cancer. *Radiology* **285**, 358–375 (2017).
21. Bleyer, A. & Welch, H. G. Effect of Three Decades of Screening Mammography on Breast-Cancer Incidence. *N Engl J Med* **367**, 1998–2005 (2012).
22. Cortazar, P. & Geyer, C. E. Pathological Complete Response in Neoadjuvant Treatment of Breast Cancer. *Ann Surg Oncol* **22**, 1441–1446 (2015).
23. Croshaw, R., Shapiro-Wright, H., Svensson, E., Erb, K. & Julian, T. Accuracy of Clinical Examination, Digital Mammogram, Ultrasound, and MRI in Determining Postneoadjuvant Pathologic Tumor Response in Operable Breast Cancer Patients. *Ann Surg Oncol* **18**, 3160–3163 (2011).
24. Berg, W. A. *et al.* Diagnostic Accuracy of Mammography, Clinical Examination, US, and MR Imaging in Preoperative Assessment of Breast Cancer. *Radiology* **233**, 830–849 (2004).
25. Baker, J. A. & Soo, M. S. Breast US: Assessment of Technical Quality and Image Interpretation. *Radiology* **223**, 229–238 (2002).
26. Rodríguez-Ruiz, A. *et al.* Detection of Breast Cancer with Mammography: Effect of an Artificial Intelligence Support System. *Radiology* **290**, 305–314 (2019).
27. Harbeck, N. *et al.* Breast cancer. *Nat Rev Dis Primers* **5**, 66 (2019).
28. Di Cosimo, S. *et al.* Mammographic density to predict response to neoadjuvant systemic breast cancer therapy. *J Cancer Res Clin Oncol* **148**, 775–781 (2022).
29. Skarping, I. *et al.* Mammographic density is a potential predictive marker of pathological response after neoadjuvant chemotherapy in breast cancer. *BMC Cancer* **19**, 1272 (2019).
30. Ataseven, B. *et al.* Impact of Multifocal or Multicentric Disease on Surgery and Locoregional, Distant and Overall Survival of 6,134 Breast Cancer Patients Treated With Neoadjuvant Chemotherapy. *Ann Surg Oncol* **22**, 1118–1127 (2015).
31. Ogier Du Terrail, J. *et al.* Federated learning for predicting histological response to neoadjuvant chemotherapy in triple-negative breast cancer. *Nat Med* **29**, 135–146 (2023).
32. Huang, Z. *et al.* Artificial intelligence reveals features associated with breast cancer neoadjuvant chemotherapy responses from multi-stain histopathologic images. *npj Precis. Onc.* **7**, 14 (2023).
33. Duanmu, H. *et al.* A spatial attention guided deep learning system for prediction of pathological complete response using breast cancer histopathology images. *Bioinformatics* **38**, 4605–4612 (2022).
34. Provenzano, E. *et al.* Standardization of pathologic evaluation and reporting of postneoadjuvant specimens in clinical trials of breast cancer: recommendations from an international working group. *Modern Pathology* **28**, 1185–1201 (2015).

REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author):

All of my comments have been sufficiently addressed. The revised manuscript is greatly improved from the original.

Reviewer #3 (Remarks to the Author):

NCOMMS-24-02259A

Summary

In this retrospective study, the authors develop and validate an explainable longitudinal multi-modal fusion model, the Multi-modal Response Prediction (MRP) system, to predict NAT response in women with breast cancer. The model was developed and tested using data from 3,352 patients with breast cancer who were treated with NAT in the Netherlands, and the model was further tested on three external datasets. The MRP is composed of two independently trained models: one that includes pre-NAT mammographic imaging and radiological, histopathological, personal, and clinical variables and one that includes longitudinal MR imaging and the aforementioned variables. The primary results are that the MRP system demonstrates promising potential in predicting pCR across various NAT scenarios and that the MRP system demonstrates favorable performance to breast radiologists in pCR prediction on both the in-house and external datasets.

Strengths

- A. This topic is of high clinical relevance and importance – if NAT response can be predicted with accuracy, then NAT regimens could be modified early on based on patient response and potential toxicities.
- B. The MRP system integrates multi-modal data from a large cohort of breast cancer patients who underwent NAT (n=3,352). The MRP system was tested on three external datasets.
- C. The authors also conducted a reader study and an error analysis to better understand discrepancies between MRP and radiologist predictions.
- D. The tables and figures are detailed and comprehensive.
- E. The authors were highly responsive to Reviewer comments. Importantly, the authors added two external datasets for validation of the MRP system.

Weaknesses

None.

Reviewer #3 (Remarks on code availability):

I was able to access the algorithm code at the following link: <https://github.com/yawwG/MRP/>.

RESPONSE TO REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author):

All of my comments have been sufficiently addressed. The revised manuscript is greatly improved from the original.

Response: We greatly appreciate the referee's positive comments, and the effort dedicated to reviewing our manuscript and responses.

Reviewer #3 (Remarks to the Author):

NCOMMS-24-02259A

Summary

In this retrospective study, the authors develop and validate an explainable longitudinal multi-modal fusion model, the Multi-modal Response Prediction (MRP) system, to predict NAT response in women with breast cancer. The model was developed and tested using data from 3,352 patients with breast cancer who were treated with NAT in the Netherlands, and the model was further tested on three external datasets. The MRP is composed of two independently trained models: one that includes pre-NAT mammographic imaging and radiological, histopathological, personal, and clinical variables and one that includes longitudinal MR imaging and the aforementioned variables. The primary results are that the MRP system demonstrates promising potential in predicting pCR across various NAT scenarios and that the MRP system demonstrates favorable performance to breast radiologists in pCR prediction on both the in-house and external datasets.

Strengths

- A. This topic is of high clinical relevance and importance – if NAT response can be predicted with accuracy, then NAT regimens could be modified early on based on patient response and potential toxicities.
- B. The MRP system integrates multi-modal data from a large cohort of breast cancer patients who underwent NAT (n=3,352). The MRP system was tested on three external datasets.
- C. The authors also conducted a reader study and an error analysis to better understand discrepancies between MRP and radiologist predictions.
- D. The tables and figures are detailed and comprehensive.

E. The authors were highly responsive to Reviewer comments. Importantly, the authors added two external datasets for validation of the MRP system.

Weaknesses

None.

Reviewer #3 (Remarks on code availability):

I was able to access the algorithm code at the following link: <https://github.com/yawwG/MRP/>.

Response: We sincerely appreciate the referee's positive comments, and the time spent reviewing our manuscript and responses.